Extending the **Hilbert-Space Quadrature Conjecture** to relativistic spacetime reveals a structural correspondence at the heart of the framework: whereas the non-relativistic Schrödinger constraint is *linear* in the clock generator and relies solely on the Born rule to perform the quadratic "squaring" operation, the relativistic mass-shell constraint ($P_\mu P^\mu - m^2c^2 = 0$) is **intrinsically quadratic** at the operator level.
This relativistic extension establishes four results that can be appended directly as **Section IX** of your manuscript:

1. **Mass-Shell Factorization & Positive-Definite Klein–Gordon Quadrature:** How the lower-bounded Hardy-space clock $\mathcal{H}_C = H^2(S^1)$ ($\widehat{N} \ge 0$) automatically selects the positive-frequency sheet of the Klein–Gordon mass-shell hyperboloid and resolves the historical Klein–Gordon negative-probability problem without an external time coordinate.
2. **The Dirac Constraint as the Operator Square Root:** How the covariant Dirac constraint acts as the algebraic square root of the Klein–Gordon quadrature, and how finite-clock Lindblad dephasing (Theorem 2) automatically damps out high-frequency spinor *Zitterbewegung* when the clock energy spread is below the pair-creation threshold ($\Delta E_C \ll 2mc^2$).
3. **Proper-Time Quadrature & Quantum Lorentz Time Dilation:** How mass-energy equivalence ($m \mapsto m + \widehat{H}_C/c^2$) couples the internal circular clock fiber S^1_C to spatial momentum $\widehat{\mathbf{P}}$, deriving both special-relativistic time dilation $\tau = \gamma^{-1} t$ and momentum-superposition clock decoherence from the joint quadrature map.

IX. Relativistic Spacetime Extension: Klein–Gordon, Dirac, and Proper-Time Quadrature

A. The Relativistic Klein–Gordon Constraint and Mass-Shell Factorization

In relativistic mechanics on (1+3)-dimensional Minkowski spacetime with metric signature $\eta_{\mu\nu} = \operatorname{diag}(+1, -1, -1, -1)$, the kinematical Hilbert space on the universal cover is

$$\mathcal{H}_U^{\text{kin}} = L^2(\mathbb{R}, dx^0) \otimes L^2(\mathbb{R}^3, d^3\mathbf{x}) \otimes \mathcal{H}_{\text{int}},$$

where $x^0 = ct$ is the temporal coordinate on the universal cover, $\widehat{P}_0 = i\hbar \frac{\partial}{\partial x^0} = -\frac{1}{c}\widehat{P}_t$ is the temporal four-momentum component, $\widehat{\mathbf{P}} = -i\hbar\nabla$ is the spatial three-momentum, and $\mathcal{H}_{\text{int}}$ encodes internal rest-frame degrees of freedom with rest-mass operator $\widehat{M} = m I_{\text{int}} + \frac{1}{c^2}\widehat{H}_{\text{int}}$.
In place of the linear Schrödinger constraint, relativistic reparameterization invariance imposes the quadratic **Klein–Gordon mass-shell constraint**:

$$\widehat{C}_{\text{KG}} \vert{}\Psi\rangle = \left(\widehat{P}_\mu \widehat{P}^\mu - \widehat{M}^2 c^2\right)\vert{}\Psi\rangle = \left(\frac{1}{c^2}\widehat{P}_t^2 \otimes I_S - I_C \otimes \frac{\widehat{H}_{\text{rel}}^2}{c^2}\right)\vert{}\Psi\rangle = 0,$$

where we define the positive-definite relativistic system energy operator on $\mathcal{H}_S = L^2(\mathbb{R}^3) \otimes \mathcal{H}_{\text{int}}$ via the spectral functional calculus:

$\widehat{H}_{\text{rel}} := +\sqrt{c^2 \widehat{\mathbf{P}}^2 + \widehat{M}^2 c^4} > 0$.
Because $\widehat{P}_t \otimes I_S$ and $I_C \otimes \widehat{H}_{\text{rel}}$ commute on $\mathcal{H}_C \otimes \mathcal{H}_S$, the quadratic Klein–Gordon constraint factorizes algebraically into positive- and negative-frequency first-order factors:
$\widehat{C}_{\text{KG}} = \frac{1}{c^2} \widehat{C}_+ \widehat{C}_-, \qquad \widehat{C}_\pm := \widehat{P}_t \otimes I_S \pm I_C \otimes \widehat{H}_{\text{rel}}$.

Theorem 3 (Hardy-Space Selection of the Positive-Frequency Mass Shell and Positive-Definite Klein–Gordon Probability)

Let $\Pi_\omega : L^2(\mathbb{R}) \to L^2(S^1)$ be the Bloch–Floquet winding quotient map of Proposition 1, and let $\mathbb{P}_+ : L^2(S^1) \to H^2(S^1)$ be the orthogonal Hardy projection onto the physical lower-bounded clock space $\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0 = \{0, 1, 2, \dots\}$ equipped with the covariant Holevo phase POVM $E(d\theta) = \frac{1}{2\pi}\vert{}\theta)(\theta\vert{}\,d\theta$. Then:

1. *The Hardy projection $\mathbb{P}_+ \otimes I_S$ annihilates the negative-frequency branch $\ker(\widehat{C}_-)$ and maps $\ker(\widehat{C}_{\text{KG}})$ isomorphically onto the positive-frequency mass-shell subspace $\mathcal{H}_+ = \ker(\widehat{C}_+^{\text{circ}})$, where*

$\widehat{C}_+^{\text{circ}} \vert{}\Psi_+\rangle = \left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_{\text{rel}}\right)\vert{}\Psi_+\rangle = 0$.

1. *The conditional system state $\vert{}\psi(\theta)\rangle = ((\theta\vert{}_C \otimes I_S)\vert{}\Psi_+\rangle$ satisfies the positive-frequency relativistic Schrödinger–Foldy equation ($t = \theta/\omega$):*

$$i\hbar \frac{\partial}{\partial t} \vert{}\psi(t)\rangle = \sqrt{c^2 \widehat{\mathbf{P}}^2 + \widehat{M}^2 c^4} \, \vert{}\psi(t)\rangle,$$

and its conditional spatial quadrature density

$$\rho_{\text{KG}}(\mathbf{x} \vert{} \theta) := \frac{J_{\Psi_+}(\theta, \vert{}\mathbf{x}\rangle\langle\mathbf{x}\vert{})}{M_{\Psi_+}(\theta)} = \vert{}\langle \mathbf{x} \vert{} \psi(\theta) \rangle\vert{}^2 \ge 0$$

is strictly nonnegative and conserved ($\int_{\mathbb{R}^3} d^3\mathbf{x} \, \rho_{\text{KG}}(\mathbf{x} \vert{} \theta) = 1$) for all clock readings $\theta \in S^1$.

Proof

Under the group-averaging projector for a quadratic constraint~\cite{Marolf1995}, $\delta(\widehat{C}_{\text{KG}}) = \frac{c^2}{2(I_C \otimes \widehat{H}_{\text{rel}})} \left[\delta(\widehat{C}_+) + \delta(\widehat{C}_-)\right]$, any distributional solution $\vert{}\Psi\rangle \in \Phi^\times$ decomposes into $\vert{}\Psi\rangle = \vert{}\Psi_+\rangle + \vert{}\Psi_-\rangle$ with $\widehat{C}_\pm \vert{}\Psi_\pm\rangle = 0$. Passing to the circle S^1 via the winding quotient map Π_ω of Proposition 1, expanding in the joint Fourier-momentum basis $\{\vert{}n\rangle_C \otimes \vert{}\mathbf{p}, k\rangle_S\}$ ($n \in \mathbb{Z}$) yields the spectral condition

$$(\hbar\omega n \mp E_{\mathbf{p},k}) c_n^{(\pm)}(\mathbf{p}, k) = 0, \qquad E_{\mathbf{p},k} = \sqrt{c^2\mathbf{p}^2 + M_k^2 c^4} > 0.$$

Since $E_{\mathbf{p},k} > 0$ is strictly positive for massive particles (and nonnegative for massless modes), the negative-frequency branch $\widehat{C}_- \vert{}\Psi_-\rangle = 0$ requires

clock mode numbers n = -E_{\mathbf{p},k}/(\hbar\omega) < 0. Applying the Hardy-space projection \mathbb{P}_+ = \sum_{n=0}^\infty \vert{}n\rangle\langle n\vert{}_C (which enforces the physical ground-state condition \widehat{N} \ge 0) therefore gives (\mathbb{P}_+ \otimes I_S)\vert{}\Psi_-\rangle = 0 identically, isolating the positive-frequency branch \vert{}\Psi_+\rangle satisfying Eq. (38).
Projecting \vert{}\Psi_+\rangle onto the Holevo clock bra (\theta\vert{}_C = \sum_{n=0}^\infty e^{-in\theta}\langle n\vert{}_C produces
\vert{}\psi(\theta)\rangle = \int d^3\mathbf{p} \sum_k \alpha_k(\mathbf{p}) \, e^{-i E_{\mathbf{p},k} \theta / (\hbar\omega)} \vert{}\mathbf{p}, k\rangle_S,
which satisfies Eq. (39) upon differentiating with respect to t = \theta/\omega. Crucially, whereas the traditional Klein–Gordon current j^0 = \frac{i\hbar}{2mc^2}(\psi^* \partial_t \psi - \psi \partial_t \psi^*) becomes negative on negative-frequency modes and fails to define a probability measure on spatial position \vert{}\mathbf{x}\rangle, conditioning on the covariant Hardy clock POVM E(d\theta) defines the spatial probability density directly through the Newton–Wigner / Foldy positive-frequency quadrature Eq. (40), which is a true Hilbert-space norm square \vert{}\psi(\mathbf{x}, \theta)\vert{}^2 \ge 0 normalized to unity for all \theta \in S^1. \blacksquare

B. The Dirac Constraint as the Algebraic Square Root and Zitterbewegung Dephasing

To incorporate spin-\frac{1}{2} fermions without taking a nonlocal square root of the Laplacian, we tensor the system space with the four-component Dirac spinor space, \mathcal{H}_S = L^2(\mathbb{R}^3) \otimes \mathbb{C}^4, endowed with Clifford matrices \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4.
The covariant first-order Dirac constraint on \mathcal{H}_C^{\text{univ}} \otimes \mathcal{H}_S is
\widehat{C}_{\text{cov}} \vert{}\Psi\rangle = \left(\gamma^0 \widehat{P}_0 + \gamma^j \widehat{P}_j - mc I_4\right)\vert{}\Psi\rangle = 0.
Multiplying on the left by c\gamma^0 (using (\gamma^0)^2 = I_4, \alpha^j := \gamma^0\gamma^j, and \beta := \gamma^0) puts the constraint into canonical bipartite form:
\widehat{C}_{\text{Dirac}} \vert{}\Psi\rangle = \left(\widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_D\right)\vert{}\Psi\rangle = 0, \qquad \widehat{H}_D := c\,\bm{\alpha}\cdot\widehat{\mathbf{P}} + \beta m c^2.
Because \widehat{H}_D^2 = (c^2\widehat{\mathbf{P}}^2 + m^2c^4)I_4 = \widehat{H}_{\text{rel}}^2 I_4, the Dirac Hamiltonian has both positive and negative energy branches \operatorname{Spec}(\widehat{H}_D) = (-\infty, -mc^2] \cup [mc^2, +\infty).

Proposition 3 (Foldy–Wouthuysen Decomposition and Finite-Clock Damping of Zitterbewegung)

Let U_{\text{FW}} = \exp\left(\frac{1}{2}\beta \frac{\bm{\alpha}\cdot\widehat{\mathbf{P}}}{\vert{}\widehat{\mathbf{P}}\vert{}} \arctan\left(\frac{\vert{}\widehat{\mathbf{P}}\vert{}}{mc}\right)\right) be the unitary Foldy–Wouthuysen transformation diagonalizing \widehat{H}_D into particle (\beta = +1) and antiparticle (\beta = -1) spinor sectors:
U_{\text{FW}} \widehat{H}_D U_{\text{FW}}^\dagger = \beta \widehat{H}_{\text{rel}} = \begin{pmatrix} +\widehat{H}_{\text{rel}} & 0 \\ 0 & -\widehat{H}_{\text{rel}} \end{pmatrix}.

If the clock energy spectrum is shifted by a reference zero-point offset $E_0 > mc^2$ so that $\vert{}\Psi\rangle$ supports a coherent superposition of positive- and negative-energy Dirac spinors $\vert{}u_+(\mathbf{p})\rangle$ and $\vert{}v_-(\mathbf{p})\rangle$, then under a finite-bandwidth clock of energy spread ΔE_C (Theorem 2), the cross-branch Zitterbewegung coherence $\rho_{+-}(\mathbf{p}, t)$ decays at the intrinsic Lindblad rate:
$\Gamma_{\text{Zitt}}(\mathbf{p}) = \frac{(E_{\mathbf{p}} - (-E_{\mathbf{p}}))^2}{2\hbar \Delta E_C} = \frac{2(c^2\mathbf{p}^2 + m^2c^4)}{\hbar \Delta E_C} \ge \frac{2 m^2 c^4}{\hbar \Delta E_C}$.

Proof

For a given momentum $\mathbf{p}$, the two energy branches of $\widehat{H}_D$ have eigenvalues $E_+ = +E_{\mathbf{p}}$ and $E_- = -E_{\mathbf{p}}$, separated by the relativistic energy gap $\Delta E_{+-} = E_+ - E_- = 2E_{\mathbf{p}} = 2\sqrt{c^2\mathbf{p}^2 + m^2c^4} \ge 2mc^2$. The Dirac velocity operator $\widehat{\mathbf{v}} = \frac{i}{\hbar}[\widehat{H}_D, \widehat{\mathbf{X}}] = c\bm{\alpha}$ mixes the positive- and negative-energy eigenspaces, producing rapid oscillatory motion (*Zitterbewegung*) of the expectation value $\langle \widehat{\mathbf{X}}(t) \rangle$ proportional to the off-diagonal coherence $\rho_{+-}(\mathbf{p}, t) = \langle u_+(\mathbf{p}) \vert{} \rho_S(t) \vert{} v_-(\mathbf{p}) \rangle e^{-i 2 E_{\mathbf{p}} t/\hbar}$.
Applying the finite-clock Lindblad master equation of Theorem 2, Eq. (20), with $E_j - E_k = 2E_{\mathbf{p}}$ and $T_C = \hbar / \Delta E_C$, the off-diagonal particle-antiparticle coherence decays exponentially with rate
$\Gamma_{+-}(\mathbf{p}) = \frac{T_C}{2\hbar^2}(2E_{\mathbf{p}})^2 = \frac{2 E_{\mathbf{p}}^2}{\hbar \Delta E_C}$,
or, in the single-shot Gaussian clock convolution of Eq. (21) with $\sigma_t = \hbar/(2\Delta E_C)$, is suppressed instantaneously by the Gaussian factor
$\exp(-\frac{1}{2}(2E_{\mathbf{p}}\sigma_t/\hbar)^2) = \exp(-E_{\mathbf{p}}^2 / (2\Delta E_C^2))$.
Thus, for any physical clock whose energy spread lies below the pair-creation threshold ($\Delta E_C \ll 2mc^2$), finite-clock quadrature resolution automatically coarse-grains away *Zitterbewegung*, projecting the observed Dirac dynamics onto the smooth Foldy–Wouthuysen positive-energy trajectory. $\blacksquare$

C. Internal Proper-Time Quadrature and Quantum Lorentz Time Dilation

Finally, we show how special-relativistic time dilation—and its quantum superposition correction—emerges directly when an internal circular clock fiber $\mathcal{H}_{C,\text{int}} = H^2(S^1)$ is carried by a moving relativistic particle.
Consider a tripartite universe $\mathcal{H}_U = \mathcal{H}_{C,\text{lab}} \otimes \mathcal{H}_{\text{cm}} \otimes \mathcal{H}_{C,\text{int}}$, where:

- $\mathcal{H}_{C,\text{lab}}$ is an inertial reference clock measuring coordinate phase $\theta_{\text{lab}} = \omega_{\text{lab}} t$,
- $\mathcal{H}_{\text{cm}} = L^2(\mathbb{R}^3)$ is the particle's center-of-mass motion with momentum $\widehat{\mathbf{P}}$, and
- $\mathcal{H}_{C,\text{int}} = H^2(S^1)$ is the particle's internal proper-time clock with Hamiltonian $\widehat{H}_{\text{int}} = \hbar\omega_0 \widehat{N}_{\text{int}}$ and Holevo phase POVM $E_{\text{int}}(d\theta_{\text{int}})$, reading proper time $\tau =$

\theta_{\text{int}} / \omega_0.

Theorem 4 (Emergent Lorentz Time Dilation and Proper-Time Entanglement Dephasing)

By Einstein mass-energy equivalence, the internal clock energy contributes to the particle's invariant mass operator as \widehat{M} = m I + \frac{\hbar\omega_0}{c^2}\widehat{N}_{\text{int}}. In the regime where the internal clock energy is small compared to the rest mass (\hbar\omega_0 \langle\widehat{N}_{\text{int}}\rangle \ll mc^2), expanding the relativistic Hamiltonian \widehat{H}_{\text{rel}} = \sqrt{c^2\widehat{\mathbf{P}}^2 + (mc^2 + \hbar\omega_0 \widehat{N}_{\text{int}})^2} to first order in \widehat{H}_{\text{int}}/(mc^2) yields:

$$\widehat{H}_{\text{rel}} = \widehat{E}_{\mathbf{P}} \otimes I_{\text{int}} + \widehat{\gamma}(\widehat{\mathbf{P}})^{-1} \otimes \hbar\omega_0 \widehat{N}_{\text{int}} + \mathcal{O}\!\left(\frac{\widehat{H}_{\text{int}}^2}{mc^2}\right),$$

where \widehat{E}_{\mathbf{P}} = \sqrt{c^2\widehat{\mathbf{P}}^2 + m^2c^4} and \widehat{\gamma}(\widehat{\mathbf{P}}) := \frac{\widehat{E}_{\mathbf{P}}}{mc^2} = \sqrt{1 + \frac{\widehat{\mathbf{P}}^2}{m^2c^2}} is the quantum Lorentz boost operator. Consequently:

1. **Classical Lorentz Dilation:** *For a state localized around momentum \mathbf{p}_0, the conditional proper-time quadrature distribution Q_\Psi(\tau \vert{} t) = \langle \Psi(t) \vert{} I_{\text{cm}} \otimes E_{\text{int}}(d\tau) \vert{} \Psi(t) \rangle peaks along the dilated proper-time trajectory*

$$\langle \tau(t) \rangle = \gamma(\mathbf{p}_0)^{-1} t = \frac{t}{\sqrt{1 + \mathbf{p}_0^2 / (m^2 c^2)}}.$$

1. **Quantum Proper-Time Dephasing in Momentum Superpositions:** *If the particle's center of mass is prepared in a coherent superposition of two relativistic momentum wavepackets \vert{}\psi_{\text{cm}}\rangle = c_1 \vert{}\mathbf{p}_1\rangle + c_2 \vert{}\mathbf{p}_2\rangle with distinct Lorentz factors \gamma_1 \neq \gamma_2, tracing out the center-of-mass momentum induces entanglement dephasing of the internal clock coherences:*

$$\vert{}\rho_{\text{int}}^{n,m}(t)\vert{} = \vert{}\rho_{\text{int}}^{n,m}(0)\vert{} \sqrt{\vert{}c_1\vert{}^4 + \vert{}c_2\vert{}^4 + 2\vert{}c_1\vert{}^2\vert{}c_2\vert{}^2 \cos\!\left[(n - m)\omega_0 \left(\gamma_1^{-1} - \gamma_2^{-1}\right) t\right]}.$$

Proof

Taylor expanding the operator square root $f(x) = \sqrt{c^2\mathbf{p}^2 + (mc^2 + x)^2}$ around $x = 0$ gives $f(0) = E_{\mathbf{p}}$ and $f'(0) = \frac{mc^2}{E_{\mathbf{p}}} = \gamma(\mathbf{p})^{-1}$, establishing Eq. (48). Under Theorem 1, conditioning on the laboratory clock reading $t = \theta_{\text{lab}}/\omega_{\text{lab}}$ yields the joint center-of-mass and internal-clock state

$$\vert{}\psi(t)\rangle = e^{-i\widehat{H}_{\text{rel}} t / \hbar} \vert{}\psi(0)\rangle = \int d^3\mathbf{p} \, e^{-i E_{\mathbf{p}} t / \hbar} \sum_{n=0}^{\infty} c_n(\mathbf{p}) \, e^{-i n \omega_0 \gamma(\mathbf{p})^{-1} t} \vert{}\mathbf{p}\rangle_{\text{cm}} \otimes \vert{}n\rangle_{\text{int}}.$$

Projecting the internal clock onto the covariant Holevo phase state $(\theta_{\text{int}}\vert{}_{\text{int}} = \sum_{n=0}^\infty e^{-in\theta_{\text{int}}}\langle$

n\vert{}_{\text{int}} with \theta_{\text{int}} = \omega_0 \tau yields the conditional amplitude \langle \mathbf{p}, \theta_{\text{int}} \vert{} \psi(t) \rangle = e^{-i E_{\mathbf{p}} t / \hbar} \sum_{n=0}^{\infty} c_n(\mathbf{p}) \, e^{i n \omega_0 \left(\tau - \gamma(\mathbf{p})^{-1} t\right)} = e^{-i E_{\mathbf{p}} t / \hbar} \, \chi_{\mathbf{p}}\!\left(\omega_0\left[\tau - \gamma(\mathbf{p})^{-1} t\right]\right).
Taking the modulus squared \vert{}\langle \mathbf{p}, \theta_{\text{int}} \vert{} \psi(t) \rangle\vert{}^2 = \vert{}\chi_{\mathbf{p}}(\omega_0[\tau - \gamma(\mathbf{p})^{-1} t])\vert{}^2 shows that the proper-time probability distribution is shifted cleanly to \tau = \gamma(\mathbf{p})^{-1} t, proving Eq. (49). Finally, for the coherent momentum superposition \vert{}\psi(0)\rangle = (c_1 \vert{}\mathbf{p}_1\rangle + c_2 \vert{}\mathbf{p}_2\rangle) \otimes \sum_n \beta_n \vert{}n\rangle_{\text{int}}, tracing out \mathcal{H}_{\text{cm}} gives the reduced internal clock matrix elements
\rho_{\text{int}}^{n,m}(t) = \beta_n \beta_m^* \left(\vert{}c_1\vert{}^2 e^{-i(n-m)\omega_0 \gamma_1^{-1} t} + \vert{}c_2\vert{}^2 e^{-i(n-m)\omega_0 \gamma_2^{-1} t}\right),
whose modulus immediately yields Eq. (50). This confirms **Consequence 3 of our original proposal**: entanglement between the clock and external degrees of freedom directly modulates the quality and coherence of emergent time. \blacksquare

Interactive Exploration: Relativistic Mass-Shell Quadrature, Lorentz Time Dilation & Zitterbewegung Damping

The interactive simulator below visualizes the three core equations of **Section IX**:

1. **Left Panel (Relativistic Mass-Shell & Hardy Projection):** Plots the relativistic dispersion hyperboloid E_{\pm}(\mathbf{p}) = \pm\sqrt{c^2\mathbf{p}^2 + m^2c^4} showing how the Hardy-space clock projection \mathbb{P}_+ (\widehat{N} \ge 0) selects the upper positive-frequency sheet \mathcal{H}_+ while suppressing the negative-frequency sheet \mathcal{H}_- unless Zitterbewegung interference is excited.
2. **Right Panel (Proper-Time Dilation & Zitterbewegung Envelope):** Compares the laboratory coordinate time oscillation against the Lorentz-dilated internal clock quadrature (\tau = \gamma^{-1} t) and displays the finite-clock Lindblad damping of the Dirac Zitterbewegung ripple at rate \Gamma_{\text{Zitt}} = 2E_{\mathbf{p}}^2 / (\hbar\Delta E_C).

Drop-In LaTeX Code for Section IX (Append Before \section{Conclusion})

You can paste the block below directly into the REVTeX 4.2 .tex document from the previous turn, right above \section{Conclusion}:

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%=========================================================================
=========
% SECTION IX: RELATIVISTIC KLEIN-GORDON, DIRAC & PROPER-TIME EXTENSION
%=========================================================================
=========
\section{Relativistic Spacetime Extension: Klein--Gordon, Dirac, and
Proper-Time Quadrature}
\label{sec:relativistic}
```

\subsection{The Klein--Gordon Mass-Shell Constraint and Hardy-Space Selection}
In $(1+3)$-dimensional Minkowski spacetime with metric $\eta_{\mu\nu} = diag(+1,-1,-1,-1)$, let the universal kinematical Hilbert space be $H_U^{kin} = L^2(R, dx^0) \otimes L^2(R^3, d^3x) \otimes H_{int}$, with temporal momentum $\widehat{P}_0 = -\frac{1}{c}\widehat{P}_t$, spatial momentum $\widehat{P} = -i\hbar\nabla$, and mass operator $\widehat{M} = mI_{int} + \frac{1}{c^2}\widehat{H}_{int}$. Relativistic reparameterization invariance imposes the quadratic Klein--Gordon mass-shell constraint:
\begin{equation}
    \widehat{C}_{\mathrm{KG}}|\Psi\rangle = \left(\widehat{P}_\mu \widehat{P}^\mu - \widehat{M}^2 c^2\right)|\Psi\rangle = \frac{1}{c^2}\widehat{C}_+ \widehat{C}_- |\Psi\rangle = 0, \qquad \widehat{C}_\pm := \widehat{P}_t \otimes I_S \pm I_C \otimes \widehat{H}_{\mathrm{rel}},
    \label{eq:kg_constraint}
\end{equation}
where $\widehat{H}_{rel} := +\sqrt{c^2\widehat{P}^2 + \widehat{M}^2 c^4} > 0$.

\begin{theorem}[Hardy-Space Selection of Positive Frequency and Positive-Definite Klein--Gordon Quadrature]
\label{thm:kg_hardy}
Under the Bloch--Floquet winding quotient $\Pi_\omega : L^2(R) \to L^2(S^1)$ followed by the orthogonal Hardy-space projection $P_+ : L^2(S^1) \to H^2(S^1)$ onto lower-bounded clock states ($\operatorname{Spec}(\widehat{N}) = \mathbb{N}_0$), the negative-frequency mass-shell sheet $\ker(\widehat{C}_-)$ is annihilated identically: $(P_+ \otimes I_S)\ker(\widehat{C}_-) = \{0\}$. The surviving physical subspace $H_+ = \ker(\hbar\omega\widehat{N} \otimes I_S - I_C \otimes \widehat{H}_{rel})$ yields a conditional wave function $|\psi(\theta)\rangle = ((\theta|_C \otimes I_S)|\Psi_+\rangle$ obeying the positive-frequency relativistic Schr\"odinger--Foldy equation ($t = \theta/\omega$),
\begin{equation}
    i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \sqrt{c^2\widehat{\mathbf{P}}^2 + \widehat{M}^2 c^4}\,|\psi(t)\rangle,
\end{equation}
with a strictly nonnegative, conserved conditional spatial quadrature density $\rho_{\mathrm{KG}}(\mathbf{x}|\theta) = |\langle\mathbf{x}|\psi(\theta)\rangle|^2 / \||\psi(\theta)\|^2 \ge 0$.
\end{theorem}

\begin{proof}
In the Fourier-momentum basis $\{|n\rangle_C \otimes

|\mathbf{p},k\rangle_S\}$, the factored constraint $\widehat{C}_\pm|\Psi_\pm\rangle = 0$ enforces $\hbar\omega n = \pm E_{p,k}$ with $E_{p,k} = \sqrt{c^2 p^2 + M_k^2 c^4} > 0$. Because the negative-frequency sheet requires $n = -E_{p,k}/(\hbar\omega) < 0$, the Hardy projector $\mathbb{P}_+ = \sum_{n=0}^\infty |n\rangle\langle n|_C$ annihilates $|\Psi_-\rangle$, isolating $|\Psi_+\rangle$ and rendering the conditional spatial probability density a positive-definite Hilbert-space norm square on H_S.
\end{proof}

\subsection{The Dirac Constraint and Finite-Clock Damping of Zitterbewegung}
For spin-$\frac{1}{2}$ fermions on $H_S = L^2(R^3) \otimes C^4$, multiplying the covariant Dirac constraint $(\gamma^\mu \widehat{P}_\mu - mc I_4)|\Psi\rangle = 0$ by $c\gamma^0$ yields
\begin{equation}
    \widehat{C}_{\mathrm{Dirac}}|\Psi\rangle = \left(\widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_D\right)|\Psi\rangle = 0, \qquad \widehat{H}_D = c\,\bm{\alpha}\cdot\widehat{\mathbf{P}} + \beta m c^2.
    \label{eq:dirac_constraint}
\end{equation}
By Theorem~\ref{thm:lindblad}, for a clock with finite energy spread ΔE_C, coherent *Zitterbewegung* oscillations between positive- and negative-energy Dirac spinor branches ($E_+ - E_- = 2E_p \geq 2mc^2$) experience intrinsic Lindblad dephasing at the rate
\begin{equation}
    \Gamma_{\mathrm{Zitt}}(\mathbf{p}) = \frac{(2E_{\mathbf{p}})^2}{2\hbar \Delta E_C} = \frac{2(c^2\mathbf{p}^2 + m^2c^4)}{\hbar \Delta E_C} \ge \frac{2m^2c^4}{\hbar \Delta E_C},
    \label{eq:zitt_damping}
\end{equation}
dynamically projecting the conditional evolution onto the smooth Foldy--Wouthuysen positive-energy sector whenever $\Delta E_C \ll 2mc^2$.

\subsection{Internal Proper-Time Quadrature and Quantum Lorentz Dilation}
When an internal Hardy clock $H_{C,int} = H^2(S^1)$ with Hamiltonian $\widehat{H}_{int} = \hbar\omega_0 \widehat{N}_{int}$ is carried by a moving particle of rest mass m, mass-energy equivalence $\widehat{M} = mI + \widehat{H}_{int}/c^2$ yields the first-order expansion
\begin{equation}
    \widehat{H}_{\mathrm{rel}} = \widehat{E}_{\mathbf{P}} \otimes I_{\mathrm{int}} + \widehat{\gamma}(\widehat{\mathbf{P}})^{-1} \otimes

\hbar\omega_0 \widehat{N}_{\mathrm{int}} +
\mathcal{O}\!\left(\frac{\widehat{H}_{\mathrm{int}}^2}{mc^2}\right),
\qquad \widehat{\gamma}(\widehat{\mathbf{P}}) = \sqrt{1 +
\frac{\widehat{\mathbf{P}}^2}{m^2 c^2}}.
    \label{eq:proper_time_hamiltonian}
\end{equation}
Conditioning on the laboratory reference clock t and reading out the internal proper-time POVM $E_{int}(d\tau)$ shifts the conditional proper-time distribution to $\langle \tau(t) \rangle = \langle \widehat{\gamma}^{-1} \rangle t$, while a center-of-mass momentum superposition $c_1|\mathbf{p}_1\rangle + c_2|\mathbf{p}_2\rangle$ modulates the internal clock coherence by the quantum twin-paradox visibility factor $\sqrt{|c_1|^4 + |c_2|^4 + 2|c_1|^2|c_2|^2 \cos[(n-m)\omega_0(\gamma_1^{-1} - \gamma_2^{-1})t]}$ ~\cite{SmithAhmadi2020}.

(If you include the citation \cite{SmithAhmadi2020} in the LaTeX bibliography, add: \bibitem{SmithAhmadi2020} A.~R.~H.~Smith and M.~Ahmadi, Quantum clocks observe classical and quantum time dilation, Nat.\ Commun.\ \textbf{11}, 5360 (2020).)